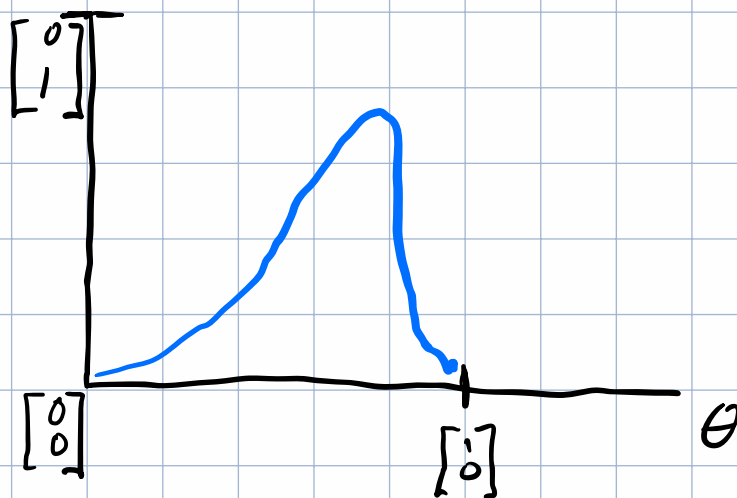


1) Without differentiating anything yet, sketch the graph of

$$L(\theta) = \theta^4 (1-\theta)^1$$

for

$$0 \leq \theta \leq 1$$



Where is the likelihood zero?

At the zeros of the polynomial: 0, 1

Roughly, where should it reach its maximum?

$\frac{4}{5}$

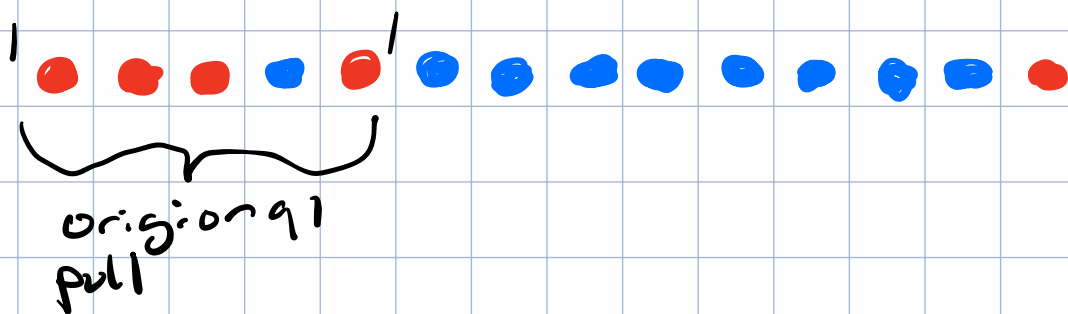
Why doesn't it peak exactly at 1?

$L(\theta) = \theta^4 (1-\theta)^1$ if you are referring to θ

If you are referring to $L(\theta)$, then the likelihood should have an area of 1 (unitary?) (not confident)

A likelihood of 1 would mean that we are certain of the value. Had we continued pulling

Balls we might have had a sequence like this



for the extended version we would have

$$L(\theta) = \theta^5 (1-\theta)^9$$

We can't be certain of any particular value because we could be in a world where we got an unlikely run. As the number of samples increases, we will become more and more certain. As samples approach ∞ , the mass of the likelihood $L(\theta)$ should be more and more centered around the real value.

Check

I know you said not to worry about the Calculus yet, but I was curious, and did not change my answers.

$$L(\theta) = \theta^4 (1-\theta)^1 = \theta^4 - \theta^5$$

find maximum where $\frac{d}{dx} L(\theta) = 0$

$$L'(\theta) = 4\theta^3 - 5\theta^4 = 0$$
$$4\theta^3 = 5\theta^4$$

$$\frac{4}{5} = \theta \quad \text{confirmed!}$$

Find the area under the curve

$$\begin{aligned}\int_0^1 L(\theta) d\theta &= \int_0^1 (\theta^4 - \theta^5) d\theta \\ &= \left. \frac{1}{4} \theta^5 - \frac{1}{5} \theta^6 + C \right|_0^1 \\ &= \frac{1}{4} - \frac{1}{5} - (0 - 0) \\ &= \frac{5}{20} - \frac{4}{20} = \frac{1}{20}\end{aligned}$$

intuition was incorrect

I was likely thinking probability distributions, but we haven't gotten there yet.